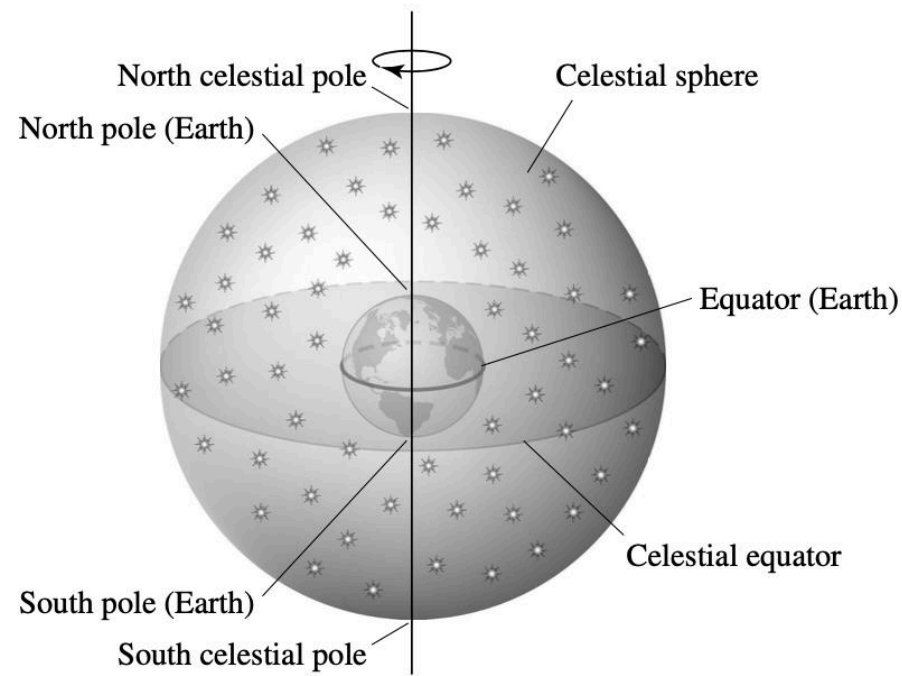


# **PH4102: Introductory Astrophysics**

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DPS and CESSI**

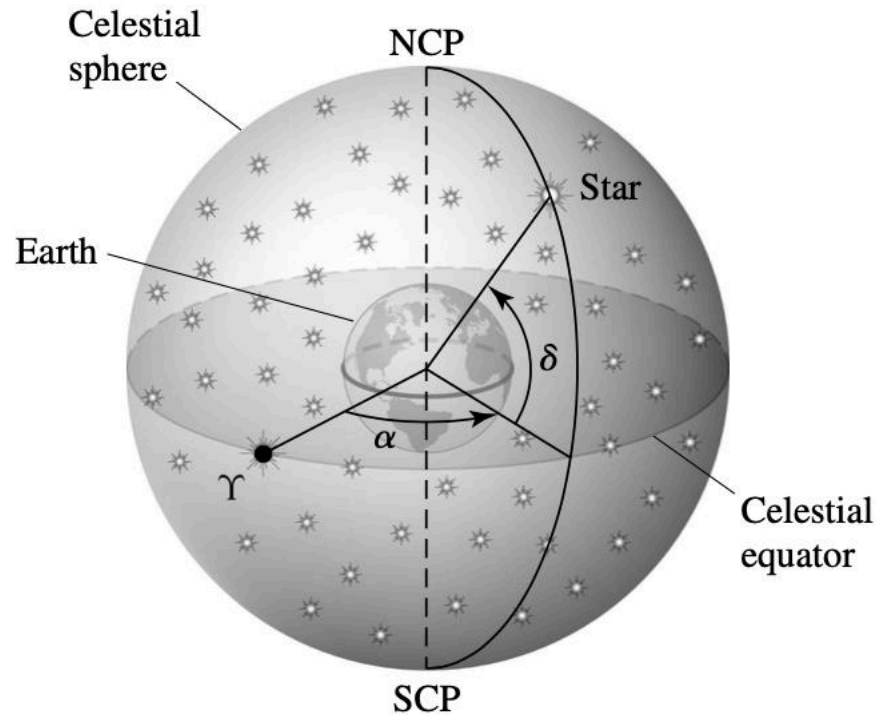
# Celestial Coordinate System



Celestial sphere is the spherical sky above us. It's a concept from geocentric universe. The stars were attached to the celestial sphere which rotated about an axis passing through north and south poles of Earth and intersecting the celestial sphere at north and south celestial poles (NCP and SCP)

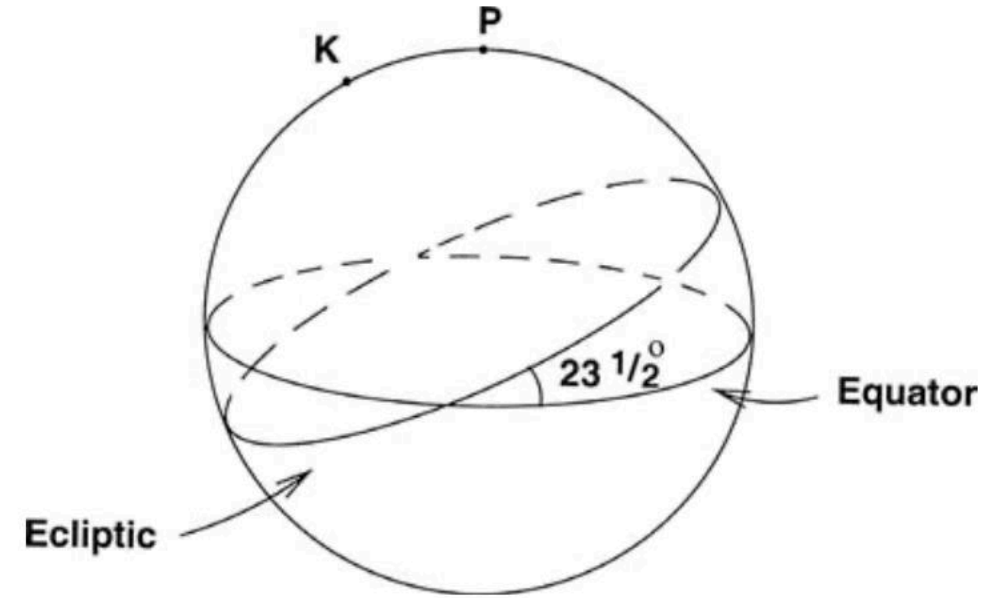
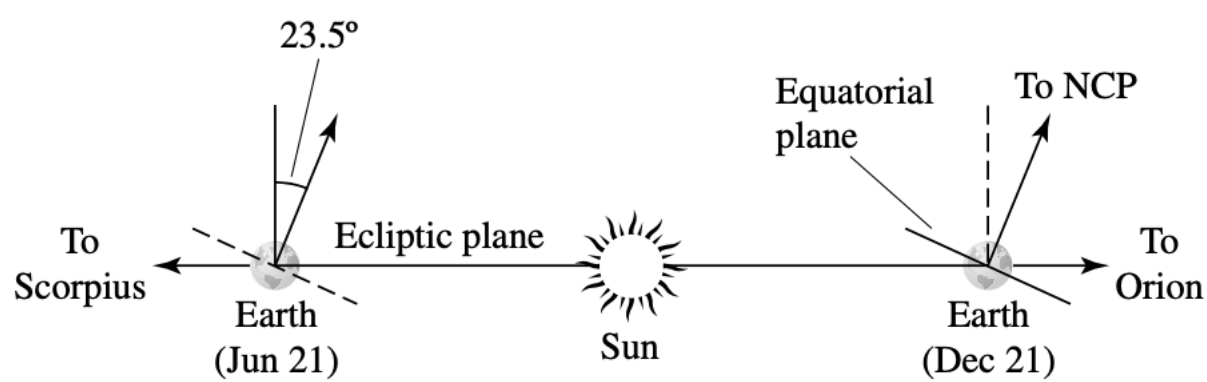
To specify a location on such 2D surface, we need two coordinates.

## The Celestial Sphere



This coordinate system is named the Equatorial coordinate system:

1. Declination ( $\delta$ ) or **Dec** is equivalent to latitude – measured in degree, north or south of the celestial equator
2. Right ascension ( $\alpha$ ) or **RA** is analogous to longitude – measured eastward along the equator from the vernal equinox  $\Upsilon$ . RA is measured in hours (rather than degree). 24 hours equivalent to 360 degree.



Earth's annual rotation about the Sun traces out a great circle in the Sky. This is **ecliptic circle**. (12 famous constellations i.e., the zodiac signs lie on the ecliptic and it appears that Sun happens to be located in different constellations during different times of year!)

Earth's equator and the ecliptic are inclined by 23.5 degree and they intersect at two points in two different times during a year: Sep 23 and Mar 20.

One of these points, lying in the constellation Aries, is taken as the zero of R.A. When the Sun is at this point, we have the **vernal equinox**.

Even with such system, there are some problem due to precession of Earth's rotation axis about an axis perpendicular to the plane of the Earth's orbit around the Sun (i.e. ecliptic plane). Precession period is 25770 years.

This causes change in the position of vernal equinox and hence RA. But note, this change happens on 25770 years timescale!

The current convention is to use the coordinates defined with respect to the positions of the poles and the equator in the year 2000: noon in Greenwich, England on January 1, 2000 (J2000.0).

## Measurement Units:

Time:

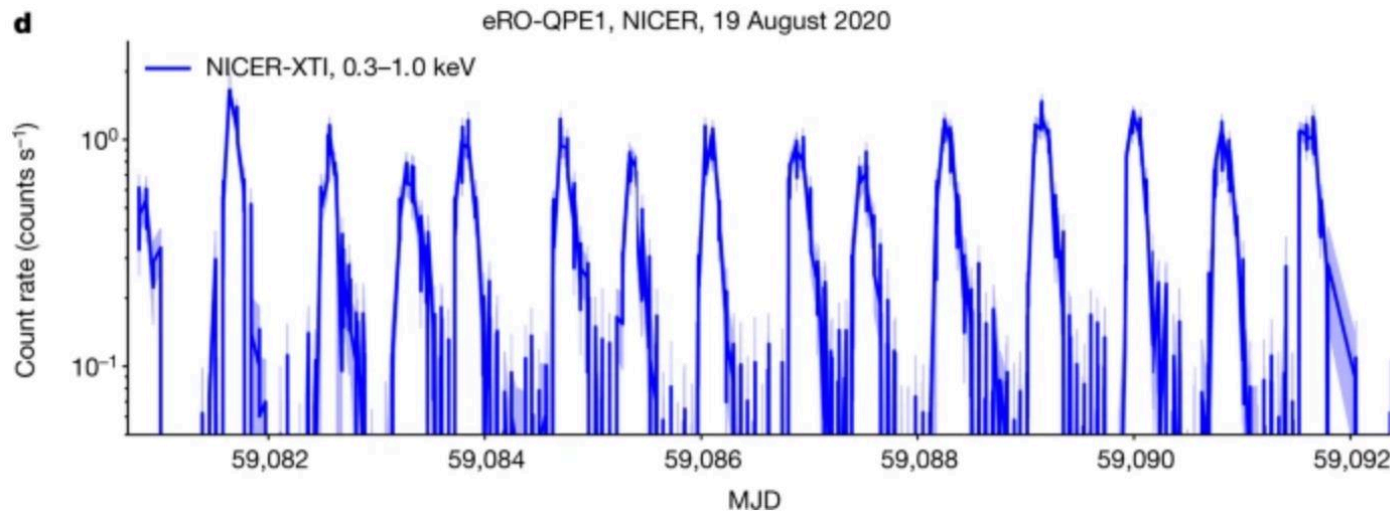
Just as the spatial coordinate, we need a universal time coordinate to mark an event in time. Our usual calendar, namely the Gregorian calendar, is of course useful.

However, many times astronomer chooses a unit system that allows **number of days count between events**. Counting based on Julian calendar is adopted.

Zero time is specified as noon on January 1, 4713 B.C. (JD 0.0). Dates (Julian Date) after this are ever increasing. J2000.0 = JD 2451545.0 (**verify it!**)

Another commonly-used designation is Modified Julian Date (MJD).  $MJD = JD - 2400000.5$ .

**Task:** Find connection between our civilian calendar and MJD.



An example showing the use of MJD on X-axis in Arcodia et al, Nature 2021. The data shows X-ray quasi periodic eruptions from two previously quiescent galaxies.

Mass:

Usually the solar mass, denoted as  $M_{\odot}$ , is taken as unit to represent mass of astrophysical objects

$$M_{\odot} = 1.99 \times 10^{30} \text{ kg}$$

However, there will be situation when kg or g may be also be used.

Length:

Light-year - the distance that light travels in one year.

$$1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$$

Another useful unit, astronomical unit (AU), which is average distance between the Sun and Earth.

$$1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$$

Another unit, parsec (parallax-second, pc).

$$1 \text{ pc} = 3.09 \times 10^{16} \text{ m}$$

pc is a measure of interstellar distances, kpc is a measure of galactic sizes, Mpc is a measure of intergalactic distances and Gpc is a measure of the visible Universe



[https://www.youtube.com/watch?v=iwIMmJs1f5o&ab\\_channel=LasCumbresObservatory](https://www.youtube.com/watch?v=iwIMmJs1f5o&ab_channel=LasCumbresObservatory)

## Distance measurement (trigonometric parallax):

Also known as surveyor's technique of triangulation.

To find the distance  $d$ , we need to observe point P from two different positions A and B separated by a known distance  $2h$ . Then, using  $d = h / \tan \theta$  we find  $d$ .

Note that  $h$  has to be sufficiently large so that  $\theta$  is much larger than the instrument error!

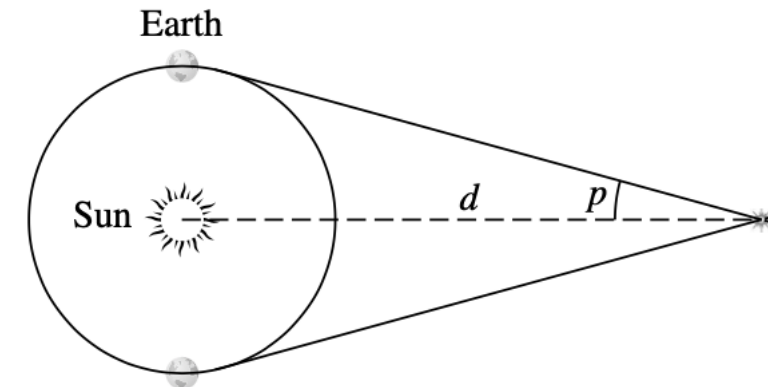
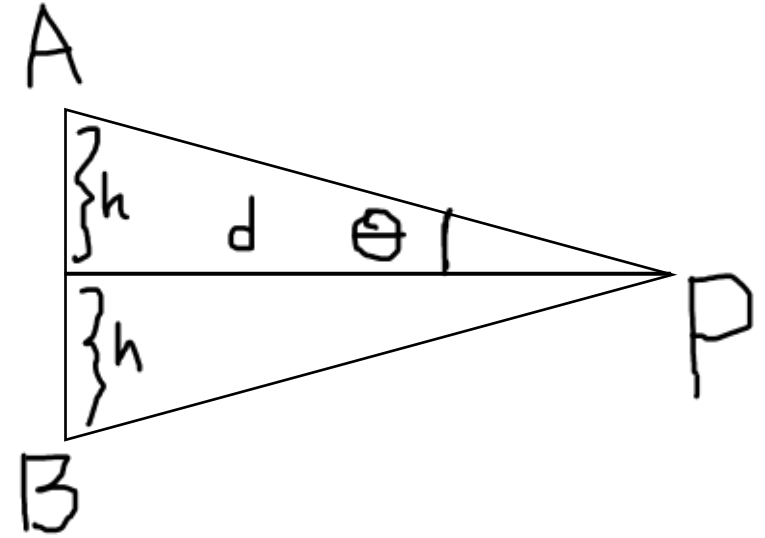
This technique can also be used to measure distance to planets by making observations from two widely separated sites on Earth.

For measuring the distance of a faraway star, we need a much larger baseline (i.e., much higher  $h$  than Earth's radius). As Earth orbits the Sun, two observations of the same star made 6 months apart employ a baseline equal to the diameter of Earth's orbit.

$$d = \frac{1 \text{ AU}}{\tan p} \simeq \frac{1}{p} \text{ AU},$$

$p$  needs to be calculated in radian. When  $p$  is 1 arcsecond, the value of  $d$  is 1 parsec.

**Task:** Putting  $p=1\text{arcsec}$ , prove to yourself  $1\text{pc} = 3.09 \times 10^{16} \text{ m}$

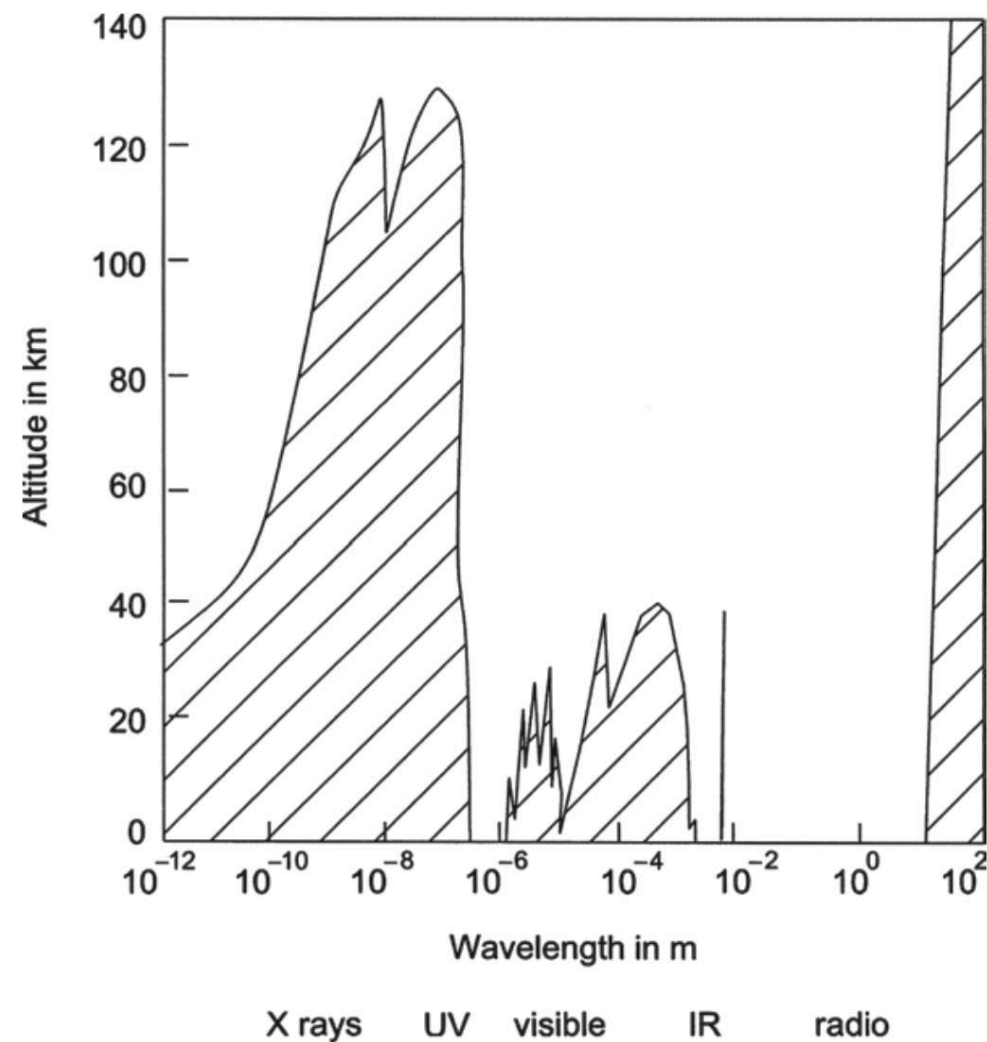


Parallax can be used to measure distance of a star which is “near”.

How about distance measurements for “far away” stars? What methods are used?

# Astronomy at different wavelenths:

Electromagnetic radiation is our main source of astronomical information. However, Earth’s atmosphere is an annoying inconvenience since the atmosphere stops significant part of the radiation bands from reaching Earth’s surface.



| Region      | Wavelength                                  |
|-------------|---------------------------------------------|
| Gamma ray   | $\lambda < 1 \text{ nm}$                    |
| X-ray       | $1 \text{ nm} < \lambda < 10 \text{ nm}$    |
| Ultraviolet | $10 \text{ nm} < \lambda < 400 \text{ nm}$  |
| Visible     | $400 \text{ nm} < \lambda < 700 \text{ nm}$ |
| Infrared    | $700 \text{ nm} < \lambda < 1 \text{ mm}$   |
| Microwave   | $1 \text{ mm} < \lambda < 10 \text{ cm}$    |
| Radio       | $10 \text{ cm} < \lambda$                   |

Optical (and IR, NIR) astronomy:

Astronomy of visible light. Modern optical astronomy started since the development of telescopes by Galileo and Newton. Galileo constructed refractive telescope in 1609 and Newton constructed reflective telescope in 1668. Majority of modern optical telescopes are reflective for following reasons:

1. a mirror is free from chromatic aberration, which affects a lens
2. making a large lens of high quality is much more difficult than making a large mirror, since a mirror requires only a defect-free surface, whereas a lens involves a volume of glass that has to be perfectly uniform and defect-free
3. a mirror can be supported from the whole of its back-side, unlike a lens which is principally supported only along its outer circumference. For a large optical component which can bend under its own weight, a proper mechanical support is crucial.

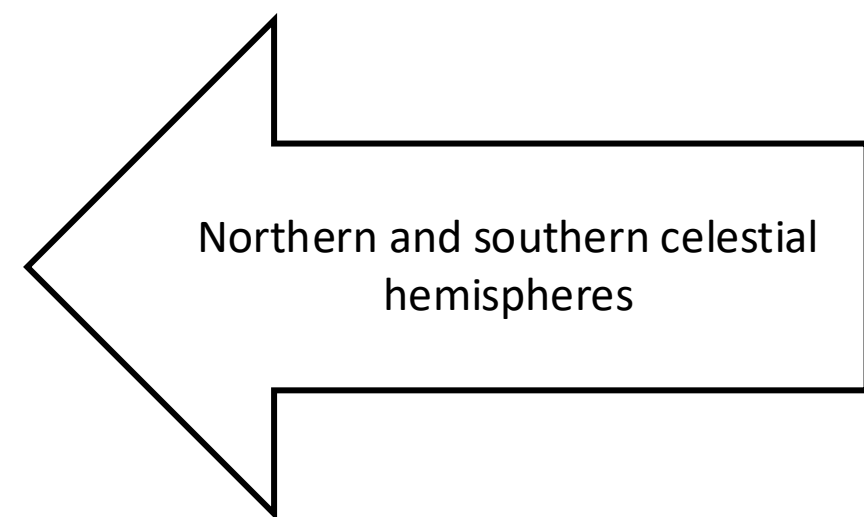
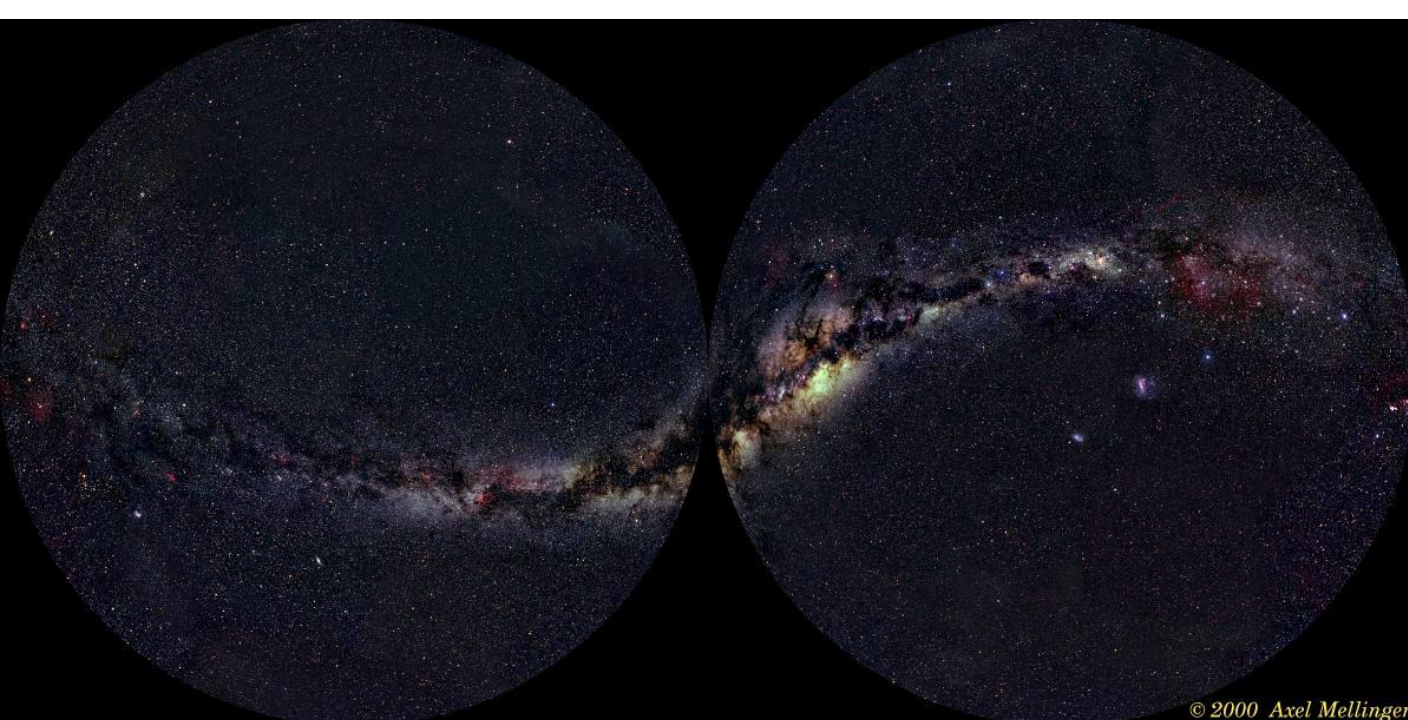
Largest diameter (around the world): Large Binocular Telescope (LBT) 11.9 m

Largest diameter (in India): Devasthal Optical telescope (DOT) 3.6 m

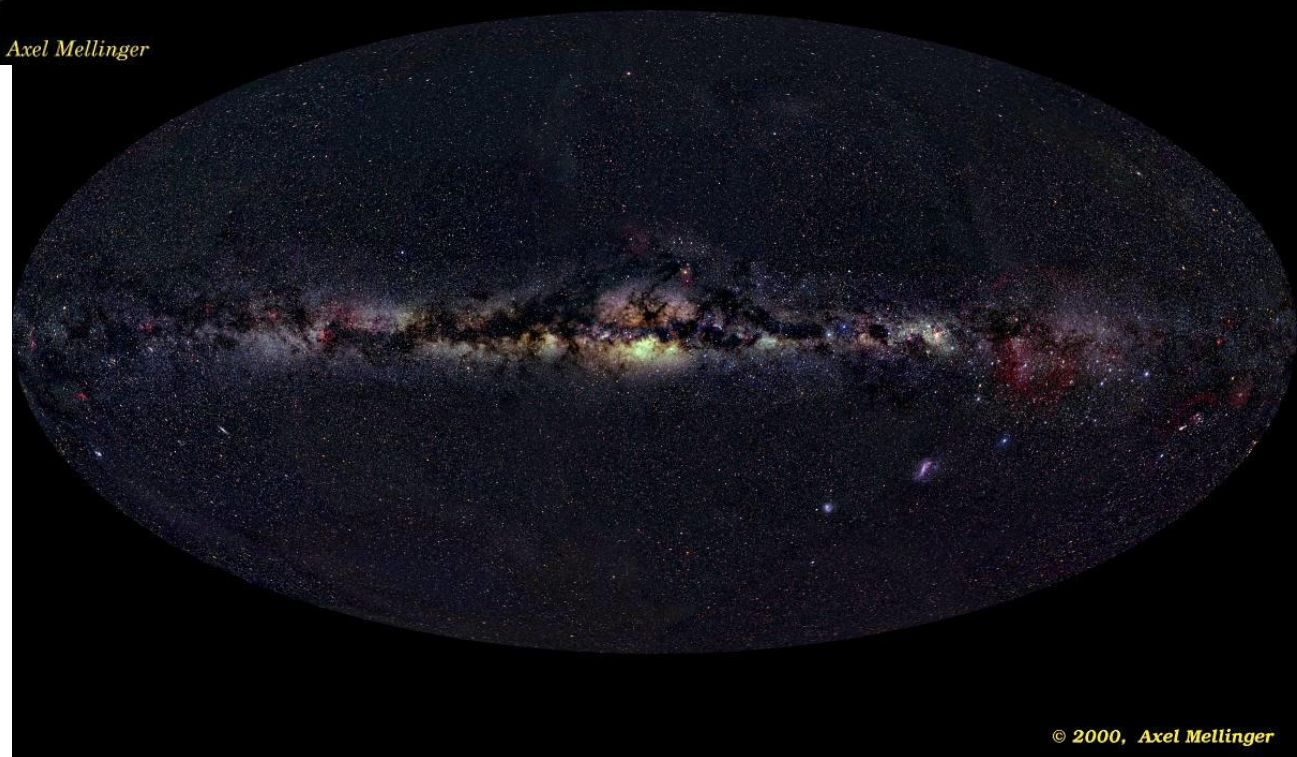
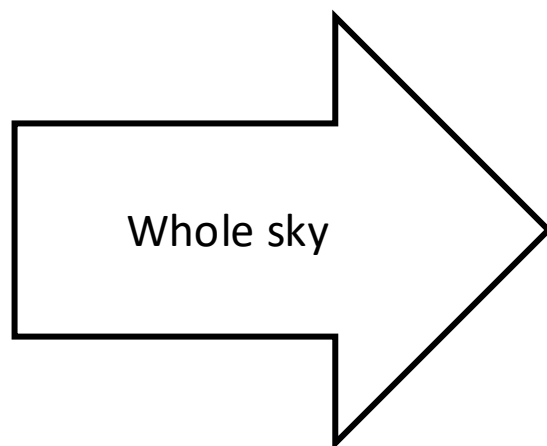
The diameter determines the resolving power of the telescope:

$$\theta = 1.22 \frac{\lambda}{D}$$

Sometimes, Earth's atmospheric effect results in blurred image. So, spaced based optical telescopes are also constructed (examples include HST, JWST etc.)



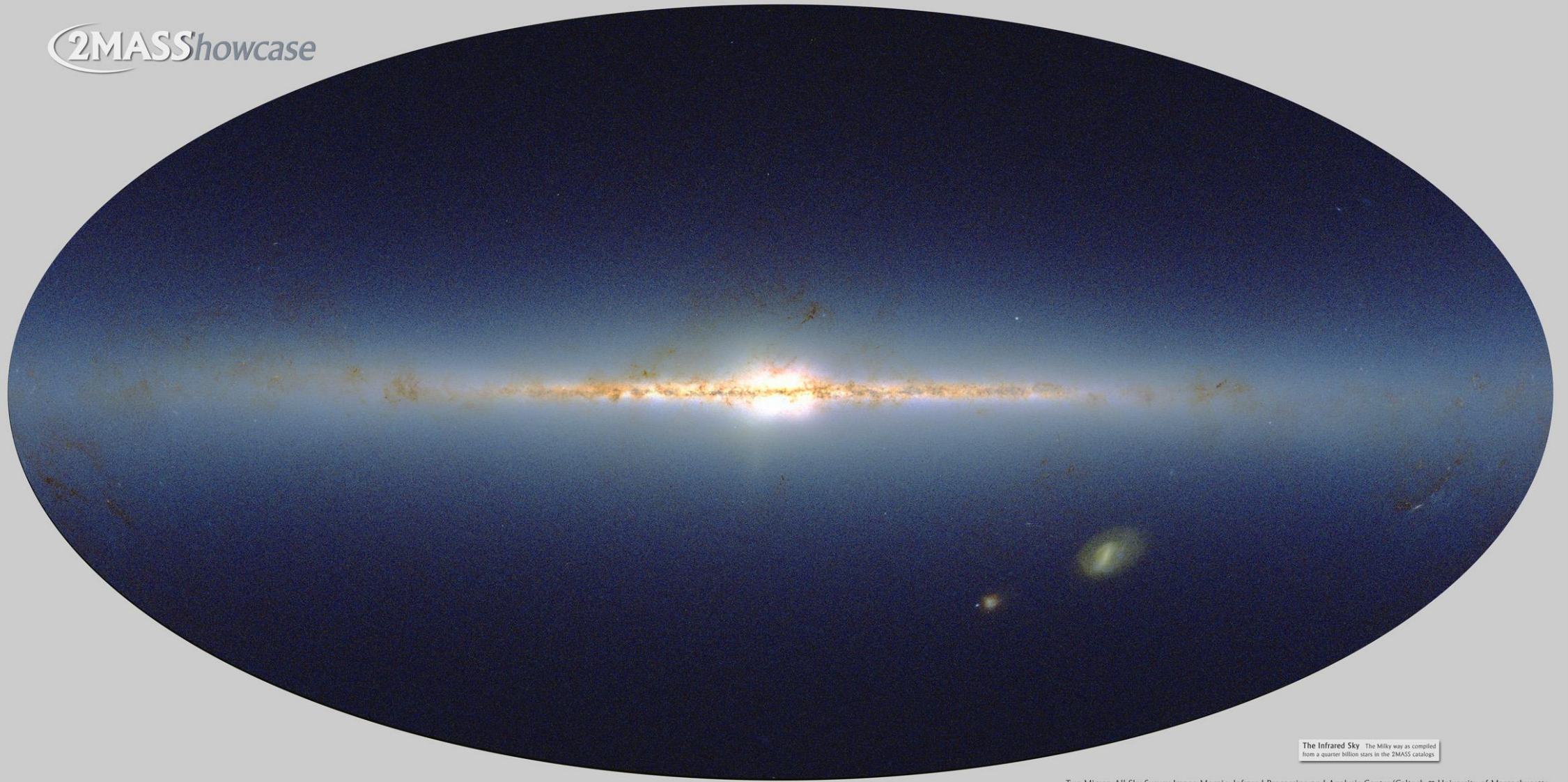
## ***The Deep Sky***



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2MASSshowcase



The Infrared Sky The Milky way as compiled from a quarter billion stars in the 2MASS catalogs

Two Micron All Sky Survey Image Mosaic: Infrared Processing and Analysis Center/Caltech & University of Massachusetts

All sky view in NIR

Radio astronomy:

It started when Jansky first detected radio signal from our galactic center in 1933.

The development of radar technology during World War II provided a major boost for the blossoming of radio astronomy after the war.

The main component of a radio telescope is an antenna in the form of a dish, which focuses the radio waves at a focal point, where receiving instruments can be kept.

Present day antenna contains several dishes suitably placed.

Some examples: Very Large Array (VLA) in New Mexico, which became operational around 1980 and consists of 27 radio antennas in a Y-shaped configuration spread over a few km.

MeerKAT in South Africa started operation in 2018 and consists of 64 antennas.

Also, Giant Metrewave Radio Telescope (GMRT) operated by NCRA-TIFR. 30 antennas.

To achieve even higher resolution, one can combine signals from different radio telescopes around the Earth operating together in a mode called the Very Long Baseline Interferometry (VLBI).

X-ray astronomy:

X-ray telescopes are placed above Earth's atmosphere.

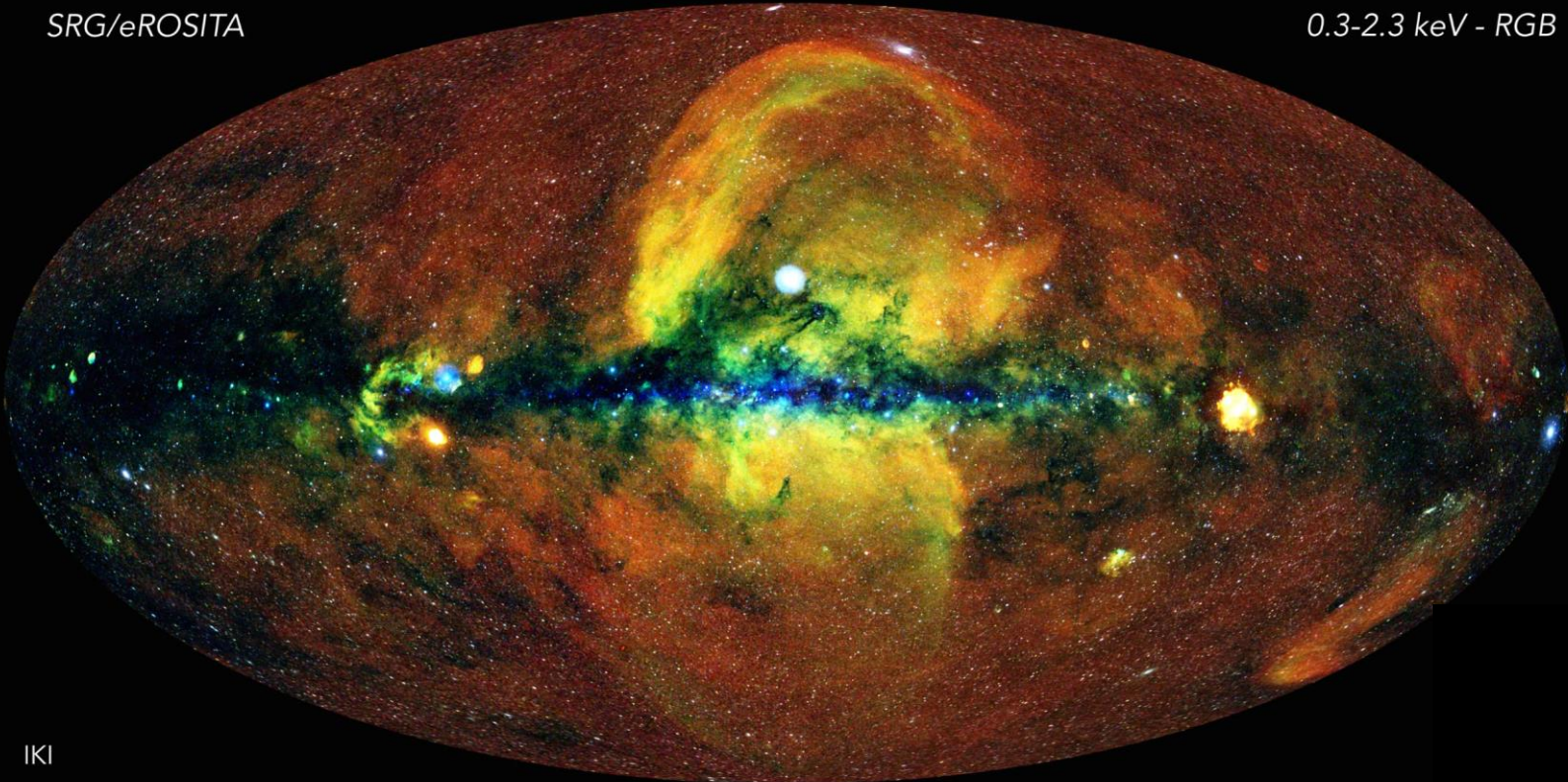
The first extraterrestrial X-ray signals were received by Geiger counters flown in a rocket by Giacconi et al., in 1962.

Later several dedicated X-ray telescopes are sent into space by various countries (majority by NASA!).

Several telescopes e.g., NuSTAR, ASTROSAT, XMM-Newton and many more are operating at different wavebands.

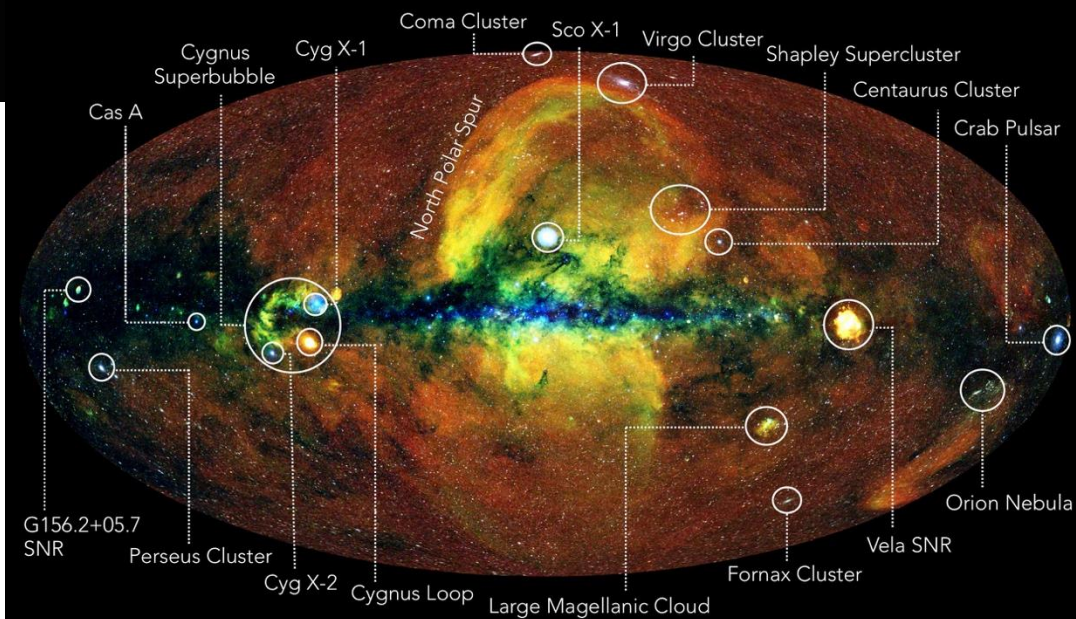
SRG/eROSITA

0.3-2.3 keV - RGB



IKI

## Navigating the eROSITA X-ray sky



IKI

SRG/eROSITA 0.3-2.3 keV - RGB Map

MPE